

# NICHOLAS SOBCHAK

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U.S. Citizen, Eligible for U.S. Security Clearance

## EXPERIENCE

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### ZipTip

Dec 2025 – Present

*Founding Software Developer*

- > Develop and implement software features using modern programming practices.
- > Collaborate with engineers using Git and GitHub for version control and code reviews.
- > Contribute to application architecture and backend development.

## EDUCATION

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### University of Maryland, Baltimore County

Aug 2025 – May 2027

Bachelor of Science in Computer Science (Expected)

### Carroll Community College

Sept 2023 – May 2025

Associate of Arts in Computer Science, GPA: 3.6 Overall, 4.0 Major

## SKILLS

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**C++:** Crow, Catch2, SQL, JSONL

**Python:** AWS Lambda, boto3, pytest, requests, JSON

**Database:** SQLite, DynamoDB

**Tools:** GitHub, Linux, Vim, Tmux, CMake, Valgrind

## PROJECTS

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### QuickQuill (C++17, Crow, SQLite)

Dec 2025 – Present

- > Designed and implemented a trie-based data structure for fast autocomplete functionality (**XXX ms**).
- > Built a RESTful backend in C++ leveraging the Crow web framework and SQLite for persistent storage and built API endpoints to serve both command-line and web-based clients.
- > Optimized database queries for very high-performance word lookup (**1000+ QPS**).
- > Implemented automated testing of core functionality using Catch2.
- > Developed a JSONL-to-SQLite data ingestion pipeline for efficient data population.

## LEADERSHIP AND ACTIVITIES

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### Ice Hockey

Team Captain; Two-time State Champion; Two-time National Qualifier

### Orchestra

Section Leader; demonstrated collaboration, discipline, and adaptability